Assignment -1

Course : Natural Language Understanding

Semester : 2nd

Year : 2026

Final report

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GitHub link :

https://github.com/MSG-1999/NLU_MahekGadiya_M25CSA011/tree/Assignment_1/Problem_4

Sports vs Politics News Classification Using Machine Learning

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1. Introduction

Text classification is an important task in Natural Language Processing (NLP) that automatically assigns categories to text documents. With the rapid increase in online news content, manual classification is inefficient and impractical. Hence, automated classification systems are widely used. This project aims to build a supervised machine learning model to classify news articles into **Sports** and **Politics**, two clearly distinct categories that are suitable for evaluating text classification techniques.

2. Data Collection

2.1 Dataset Source

- The dataset used in this project is the **HuffPost News Category Dataset**, which is publicly available on Kaggle. This dataset contains news articles published by HuffPost between 2012 and 2022. The dataset is provided in **JSON Lines format**, where each line represents one news article as a JSON object. The dataset was downloaded manually from Kaggle and stored locally for processing.
- Each article includes multiple fields such as:
  a) category
  b) headline
  c) short_description
  d) authors
  e) date
- Link : https://www.kaggle.com/datasets/rmisra/news-category-dataset?resource=download

2.2 Category Selection

- The original dataset contains multiple news categories. For this task, only two categories were selected:
  - **SPORTS**
  - **POLITICS**

- All other categories were discarded. This filtering step ensures that the problem remains a binary classification task, simplifying analysis while still being non-trivial.

3. Dataset Description and Analysis

3.1 Original Dataset Overview

The dataset used for this task is the News Category Dataset, which contains articles from multiple news categories.

After loading the dataset:

- Total rows in original dataset: 209,527

Since this project focuses only on binary classification between **Sports** and **Politics**, the dataset was filtered to retain only these two categories.

3.1.1 Filtering and Class Distribution

After filtering for the required categories:

- Total filtered samples: 40,679

The class distribution before balancing was:

Category	Number of Articles
POLITICS	35,602
SPORTS	5,077

This clearly shows a **severe class imbalance**, where Politics articles significantly outnumber Sports articles (approximately 7:1 ratio).

Such imbalance can bias machine learning models toward the majority class, resulting in poor generalization for the minority class.

3.1.2 Handling Class Imbalance

To address this issue, **random undersampling** was applied to the majority class (Politics). A subset of 5,077 Politics articles was randomly selected to match the number of Sports articles.

Final balanced dataset statistics:

- **Sports:** 5,077 articles
- **Politics:** 5,077 articles
- **Total samples:** 10,154 articles

This balanced dataset ensures that:

- Both classes contribute equally during training

- The classifier does not become biased toward Politics
- Evaluation metrics such as Precision, Recall, and F1-score are more meaningful

3.1.3 Text Construction

To create meaningful textual input, the following fields were combined:

- headline
- short_description

This combination provides both a concise summary and additional context for each article.

3.1.4 Label Encoding

The categorical labels were converted into numerical form for machine learning:

- SPORTS → 0
- POLITICS → 1

This encoding allows classifiers to process labels efficiently.

4. Data Preprocessing

Text data is inherently noisy. Although deep preprocessing was not explicitly applied, the following steps were handled implicitly during feature extraction:

- Conversion to lowercase.
- Removal of English stopwords.
- Tokenization.
- Removal of punctuation and irrelevant symbols.

Advanced preprocessing techniques such as stemming or lemmatization were not used, as TF-IDF with linear classifiers typically performs well without aggressive normalization.

5. Feature Representation

5.1 TF-IDF Representation

Text documents were transformed into numerical vectors using Term Frequency Inverse Document Frequency (TF-IDF). TF-IDF assigns higher weights to words that are important to a document but less frequent across the corpus.

5.2 N-grams

To capture short phrases and contextual information, both unigrams and bigrams were used. This allows the model to learn patterns such as:

- “world cup”
- “election results”
- “government policy”

The TF-IDF vectorizer was trained only on the training set to prevent data leakage

6. Experimental Setup

6.1 Dataset Split

The dataset was split ratio :

- **70% Training**
- **20% Validation**
- **10% Test**

The validation set was used for model selection, while the test set was reserved for final evaluation.

6.2 Machine Learning Models

Three machine learning techniques were implemented and compared:

A. **Multinomial Naive Bayes :**
A probabilistic classifier based on Bayes’ theorem. It is simple, fast, and commonly used as a baseline for text classification.

B. **Logistic Regression :**
A discriminative linear classifier that models the probability of class membership. It performs well on high-dimensional sparse data.

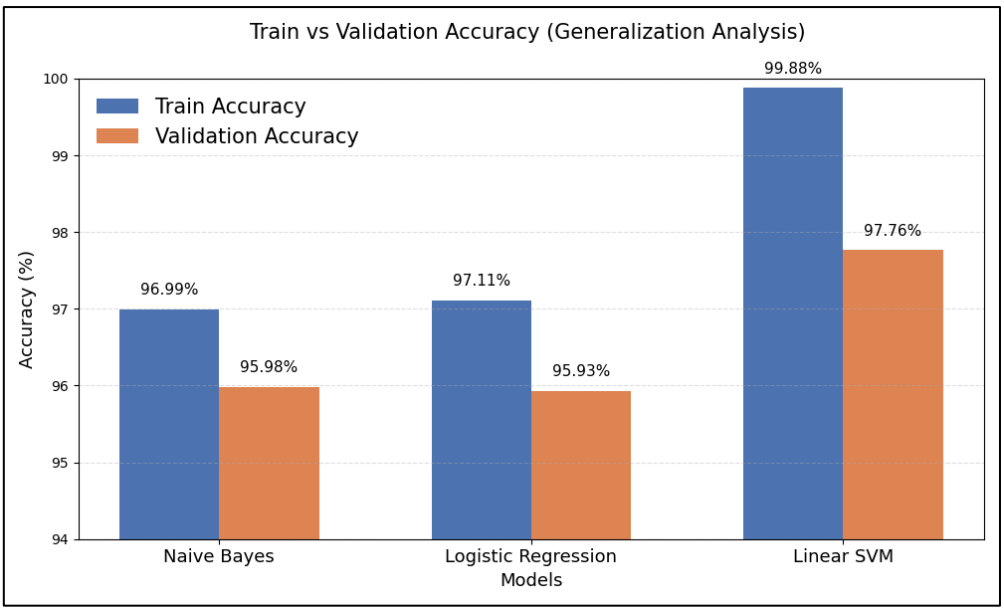
C. **Linear Support Vector Machine (SVM):**
A margin-based classifier that finds an optimal separating hyperplane. Linear SVMs are particularly effective for text classification tasks.

7. Results

7.1 (Train & validation) Accuracy :

Model	Train Accuracy	Validation Accuracy
Naive Bayes	96.99 %	95.98 %
Logistic Regression	97.11%	95.93%
Linear SVM	99.88%	97.76%

Linear SVM achieved the highest validation accuracy and showed a small gap between training and validation accuracy, indicating good generalizatio

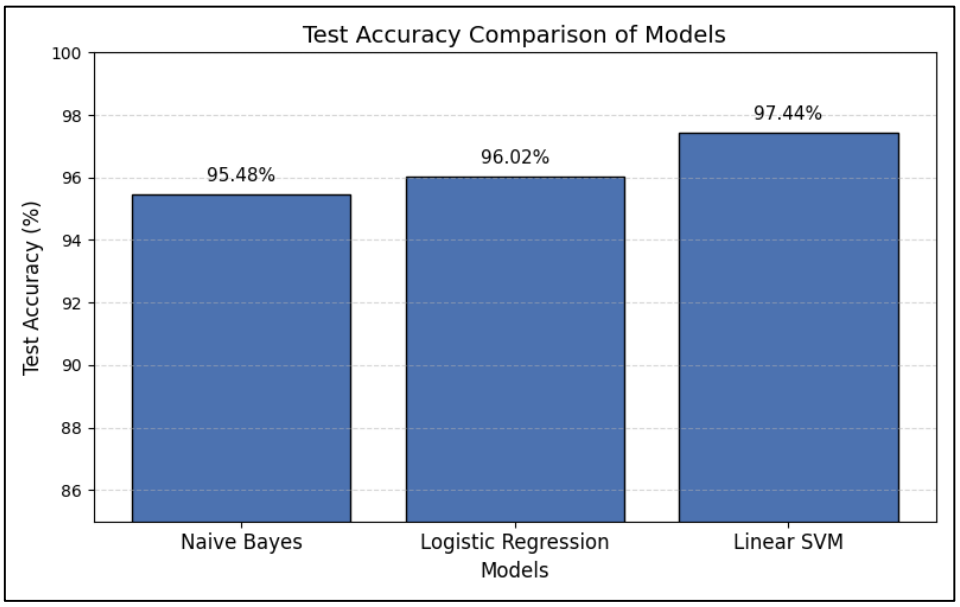


Fig(A). Train vs Validation Accuracy Plot

7.2 Test Accuracy :

Model	Test Accuracy
Naive Bayes	95.48 %
Logistic Regression	96.02 %
Linear SVM	97.44 %

The consistency between validation and test performance confirms that Linear SVM generalizes well to unseen data.

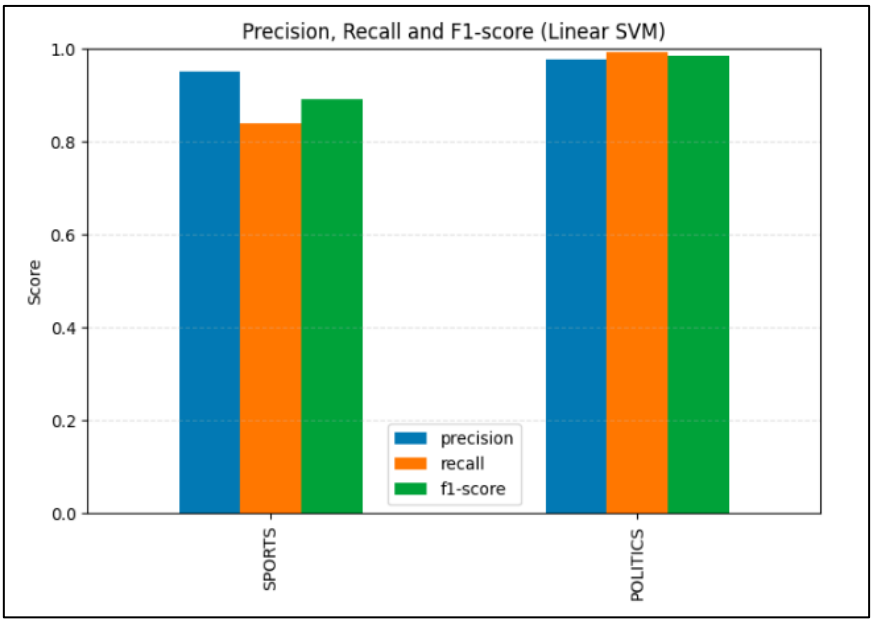


Fig(B). Test Accuracy Plot

7.3 Classification Report

The final Linear SVM model was evaluated using precision, recall, and F1-score. The results showed balanced performance across both classes, with minimal misclassification. Normalized confusion matrices further confirmed strong diagonal dominance.

Class	Precision	Recall	F1-score	Support
Sports	0.95	0.84	0.89	508
Politics	0.98	0.99	0.99	3560
Accuracy	—	—	0.97	4068
Macro Avg	0.96	0.92	0.94	4068
Weighted Avg	0.97	0.97	0.97	4068



7.4 Confusion Matrix

To analyze the classification performance in detail, normalized confusion matrices were plotted for all three models: Naive Bayes, Logistic Regression, and Linear SVM.

A confusion matrix provides insight into how well the model classifies each class by showing:

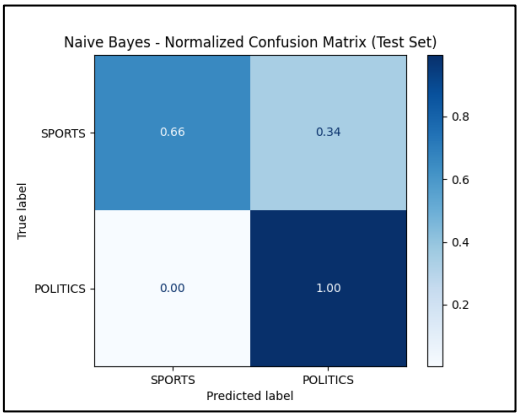
- True Positives (correct predictions)
- False Positives (incorrect positive predictions)
- False Negatives (missed predictions)
- True Negatives (correct negative predictions)

7.4.1 Naive Bayes

From the confusion matrix:

- 66% of Sports articles were correctly classified.
- 34% of Sports articles were misclassified as Politics.
- 100% of Politics articles were correctly classified.

This indicates that Naive Bayes struggles more with identifying Sports articles and shows a bias toward predicting Politics.



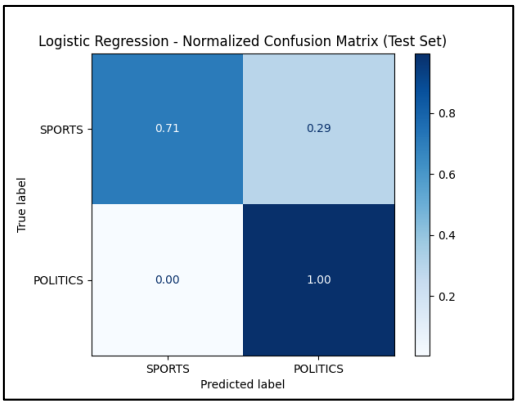
Fig(D). Naive Bayes Confusion Matrix

7.4.2 Logistic Regression

From the confusion matrix:

- 71% of Sports articles were correctly classified.
- 29% were misclassified as Politics.
- 100% of Politics articles were correctly identified.

Logistic Regression improves Sports classification compared to Naive Bayes but still misclassifies a noticeable portion of Sports article



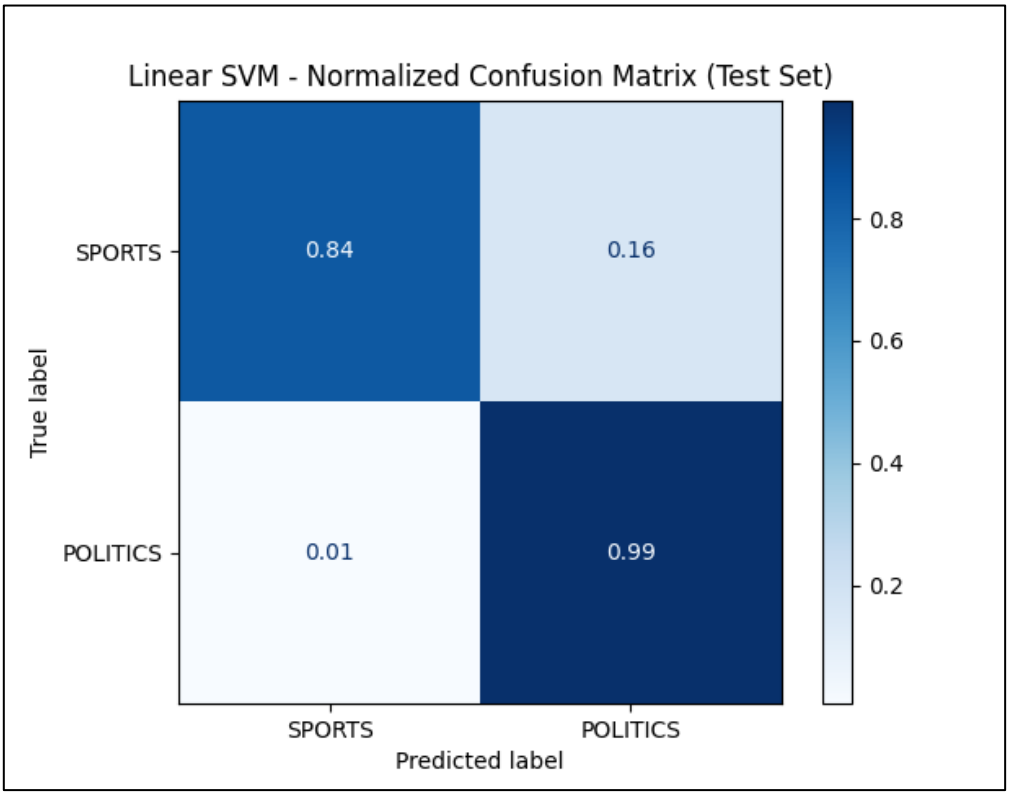
Fig(E). Logistic Regression Confusion Matrix

7.4.3 Linear SVM

From the confusion matrix:

- 84% of Sports articles were correctly classified.
- Only 16% were misclassified.
- 99% of Politics articles were correctly identified.

Linear SVM shows the strongest diagonal dominance, meaning it achieves the best balance between both classes and minimizes misclassification.



Fig(C). Linear SVM Confusion Matrix

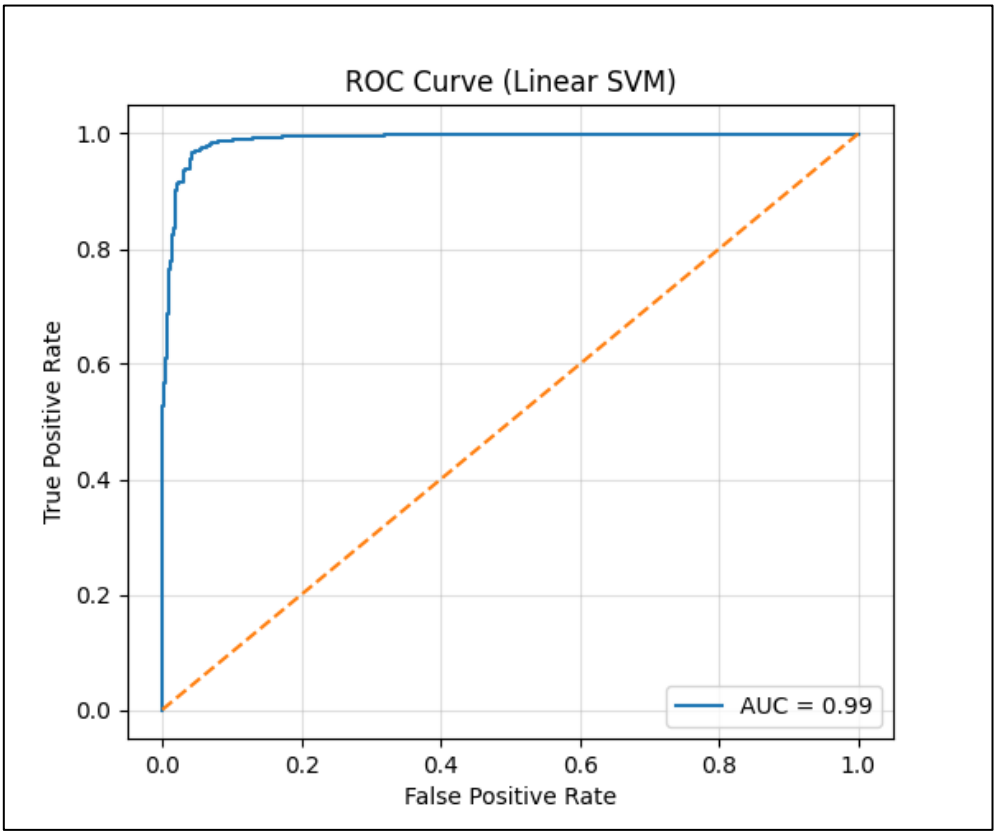
- Among the three models, Linear SVM demonstrates the most balanced and accurate classification performance. It significantly reduces misclassification of Sports articles while maintaining near-perfect prediction of Politics articles.

- The confusion matrix analysis confirms the superiority of Linear SVM observed in accuracy

7.4 ROC Curve Analysis

- To further evaluate the classification performance, the Receiver Operating Characteristic (ROC) curve was plotted for the final Linear SVM model using the test dataset.

- The ROC curve represents the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) at different classification thresholds. It provides a threshold-independent evaluation of the model's ability to distinguish between the two classes.
- The Linear SVM model achieved an Area Under the Curve (AUC) value of 0.99, which indicates excellent discriminative capability. Since an AUC value of 0.5 corresponds to random guessing and a value of 1.0 represents a perfect classifier, an AUC of 0.99 demonstrates that the model can almost perfectly separate Sports and Politics articles.
- The ROC curve lies significantly above the diagonal reference line (random classifier), confirming that the model performs substantially better than chance. The curve's proximity to the top-left corner indicates a high true positive rate with a very low false positive rate.
- Overall, the ROC analysis strengthens the conclusion that Linear SVM is highly effective for this binary text classification task and complements the high accuracy, precision, recall, and F1-score observed in previous evaluations



Fig(F). ROC Curve

8. Model Saving and Reusability

The trained Linear SVM model and the TF-IDF vectorizer were saved using the joblib library. Saving the model enables:

- Reuse without retraining
- Faster inference on new data
- Reproducibility of results

Both the model and vectorizer are required to ensure consistent feature representation.

9. Limitations of the System

Despite good performance, the system has several limitations:

- It supports only binary classification
- It works only for English news articles
- It relies on traditional bag-of-words features
- It may not generalize well to informal or noisy text
- It does not capture deep semantic relationships

11. Conclusion

This project presents a complete machine learning pipeline for classifying news articles into Sports and Politics categories. Through systematic experimentation and evaluation, Linear SVM was identified as the best-performing model. The results demonstrate that classical machine learning methods combined with appropriate feature engineering remain highly effective for text classification tasks